HARSH VEKARIYA

AI/ML Developer|Python|LLMSystems|ProductionMLEngineering

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PROFESSIONAL SUMMARY

AI/ML Developer specializing in production-scale machine learning systems and intelligent automation. Engineered enterprise CSR chatbot serving 15,000+ products with **650% response time reduction and 93% accuracy**. Built end-to-end ML pipelines processing **1,000+ daily orders** through fine-tuned CNN models. Expert in LLM integration, graph databases, and scalable backend systems using Python, TensorFlow, LangGraph, and Neo4j. Proven ability to deliver measurable business impact through intelligent automation and data-driven solutions.

TECHNICAL SKILLS

Programming & Frameworks: Python, Flask, Django, JavaScript, HTML/CSS, C/C++, Jinja2 Templating

AI/ML & Deep Learning: LangChain, LangGraph, TensorFlow/Keras, NLP, Computer Vision, CNN, Transfer Learning, Fine-Tuning, LLM Integration, Prompt Engineering, RAG Systems

Databases & Data Systems: Neo4j (Graph DB), MySQL, Vector Indexing, Full-Text Search, GraphDB, DBeaver, SQL

APIs & Web Development: RESTful APIs, Flask API, Real-Time Data Synchronization, Web Scraping, Celery, Redis Caching

DevOps & Tools: Git/GitHub, Streamlit, Flask-Admin, Flask-Login, Bootstrap, API Optimization, Model Deployment

Core Competencies: Data Pipeline Architecture, Database Modeling, Multi-Source Integration, Production ML Systems, Real-Time Analytics

ADDITIONAL STRENGTHS

Fast Learner · Problem-Solving · Time Management · Adaptability · Team Collaboration · Cross-Functional Communication · Technical Documentation

PROFESSIONAL EXPERIENCE

AI/ML DEVELOPER | Enterprise Customer Service Platform | Team: 45-50 Members

AI-PoweredCSRAgentwithMulti-SourceDataPipeline | Slack&ZendeskIntegration

Business Impact:

- Achieved **650% reduction** in customer response time through intelligent automation
- Delivered **93% accuracy** in query resolution across 15,000+ products Scaled
- enterprise-level customer service operations with measurable ROI

Data Pipeline Engineering:

- Architected and implemented automated multi-source data pipeline synchronizing Admin Dashboard APIs, vendor website scraping, and Slack-integrated CSV delivery data
- Designed real-time data flow ensuring seamless propagation of product updates, vendor metadata, and logistics information to CSR agent
- Built web scraping connectors for automated vendor site metadata extraction and synchronization

Database Architecture & Modeling:

- Engineered Neo4j graph database schema with automated relationship and node creation for 15,000+ product entities
- Implemented hybrid indexing strategy combining vector similarity search and full-text context indexing for optimal retrieval performance
- Designed dual-database architecture integrating Neo4j (product knowledge graph) with MySQL (transactional order data)

API Integration & Orchestration:

- Integrated Admin Dashboard CRUD APIs enabling real-time product management synchronization
- Developed Slack API integration for delivery data ingestion and bi-directional customer interaction workflows
- Built RESTful API endpoints ensuring seamless communication across multiple data sources

Retrieval System & Prompt Engineering:

- Designed LangGraph-based conditional routing system for intelligent query classification and Cypher query generation
- Developed prompt engineering framework for graph-based retrieval and MySQL agent coordination
- Implemented context-aware retrieval logic ensuring accurate responses across product catalog, orders, and delivery queries

Technical Stack: *Python, Neo4j,n8n,Qdrant, MySQL, LangGraph, LangChain, Vector Embeddings, REST APIs, Slack SDK, Web Scraping, Zendesk API*

KEY PROJECTS

Fine-Tuned CNN Model with Flask API for Custom Design Recognition & Order Automation

- Developed and deployed fine-tuned **ResNet50-based CNN model** classifying customer-submitted design templates, automating processing of **1,000+ custom product orders daily** with high accuracy
- Leveraged **transfer learning** with TensorFlow/Keras, customizing top layers of ResNet50 for domain-specific image classification
- Enhanced model performance using **data augmentation, early stopping, and learning rate scheduling** to ensure robustness and prevent overfitting
- Built and deployed **Flask REST API** serving real-time predictions, enabling seamless backend integration with order management and production systems
- Integrated model into web-based UI for support and operations teams, providing quick design validation and **reducing manual workload by 70%**
- Achieved significant improvements in order fulfillment speed, reduced error rates in template identification, and increased operational scalability

Skills: *CNN · ResNet50 · Fine-Tuning · Image Classification · TensorFlow/Keras · Flask API · Python · Deep Learning · Data Augmentation · Model Deployment · Order Automation*

Live Sports Score & Analytics Web App

- Built **real-time sports analytics platform** using Flask, delivering live scores, team stats, and head-to-head comparisons in interactive and scalable web interface
- Developed responsive front-end using **Jinja2 templates and Bootstrap**, ensuring seamless user experience across devices
- Integrated **Flask-Admin** for internal dashboard management and **Flask-Login** for secure authentication and session handling
- Implemented **Celery with background workers** to fetch and update live sports data asynchronously, ensuring real-time accuracy without blocking main application
- Applied **API call optimization and Redis caching strategies** to minimize latency and improve performance under high concurrent load
- Designed architecture supporting scalable user base and simultaneous live match tracking
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Skills: *Flask · Jinja2 · Bootstrap · Celery · Flask-Admin · Flask-Login · REST APIs · Redis Caching · Python · Real-Time Data · Web Development · Asynchronous Processing*

EDUCATION

BACHELOR OF ENGINEERING IN INFORMATION TECHNOLOGY
CharutarVidyaMandalUniversity | Anand,Gujarat | **GPA:8.25/10** | Oct2021 - May 2025

HSC (GSHEB)
AshadeepSchool 4 | Surat, Gujarat | **Percentage: 80.38%** | March 2020 - May 2021

SSC (GSEB)
AshadeepSchool 2 | Surat, Gujarat | **Percentage: 87.38%** | March 2019 - May 2020